PROMPT 05 | One-Page Research Problem Brief

AI Tools for Research · IIIT Allahabad · 28 October 2026 · Provisional planning; no experiments

Diagnosing Capability and Specification Uncertainty in Tool Retrieval: An Evidence-Grounded Evaluation

Motivation / prior art. Tool awareness and retrieval (MetaTool, ToolRet), capability-rich descriptions (Tool-REX, MFTR), multi-valid labeling (ToolEX), and abstention/clarification/calibration (AgentAbstain, SAGE-Agent, Zhao et al.) are already studied. The proposed distinction is **not proven novel**.

Research problem

For each user request and available tool inventory, distinguish **SELECT** (at least one justified valid tool), **NO_VALID_TOOL** (inventory lacks required capability) and **NEEDS_CLARIFICATION** (unspecified user constraint changes which available tool fits). Track multiple valid tools as accepted answers, not a fourth action. This is retrieval-stage decision evaluation, not executable agent performance.

Questions and falsifiable expectations

RQ1 (primary): Does explicit requirement/capability evidence improve three-way decisions over development-calibrated retrieval confidence? **H1 (proposed):** better held-out macro-F1 and/or decision-risk trade-off without unacceptable false selection. **Refute:** calibrated baseline matches/exceeds it or false selections increase.

RQ2: Do capability-removal versus user-specification counterfactuals produce more appropriate action transitions under the evidence policy? **Refute:** no paired transition gain. **Optional RQ3:** Do adjudicated multiple-valid tools materially change measured retrieval correctness?

Proposed design, comparators and outcomes

Inputs: independently adjudicated request–inventory pairs, tool descriptions/schemas, requirement evidence and one-factor counterfactual pairs. **Variables:** decision policy; true decision state; counterfactual type. **Core comparators:** forced top-1 (diagnostic), maximum-score/margin threshold, calibrated threshold (dev-only), minimal evidence-rule policy. Strong compatible published comparator desirable but access unknown.

Measures: decision macro-F1, false-selection rate on NO_VALID_TOOL, inappropriate clarification, paired decision consistency, multi-positive Recall@k; risk–coverage and latency if feasible. **Success:** pre-register paired improvement with confidence interval and safety guardrails; no unjustified numerical target. Negative/inconclusive findings must be reported.

Critical unknowns and feasibility gates

UNKNOWN: availability and reliability of three-way labels, valid counterfactuals, complete positive sets, licensing, matching published comparators, actual free-Colab resources and novelty. Prior Colab observation: 20,614 MetaTool rows (Query, Tool) and 199 descriptions; repository count differs. This does not validate new decision labels. Require annotation rubric, independent adjudication, leakage-safe split and calibration/test separation before freezing.

Potential contribution and selected references

INFERRED: an evidence-grounded decision-boundary diagnostic and counterfactual evaluation, with incremental value tested against calibrated baselines; not a claimed new model. **Sources:** MetaTool (Huang et al., 2024); ToolRet (Shi et al., 2025); Tool-REX (Lu et al., 2026); MFTR (Tang et al., 2026); ToolEX (2026 preprint); AgentAbstain (2026 preprint); SAGE-Agent (Findings ACL 2026); Zhao et al. (Findings EMNLP 2026). URLs and full source scope in supporting Markdown.

Readiness: CONDITIONAL GO for Prompt 06 feasibility checks only. No protocol frozen; no results, implementation or final researcher approval.